

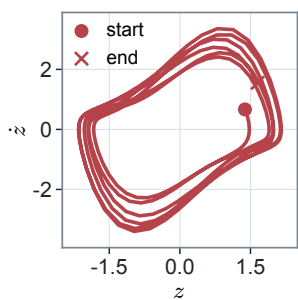
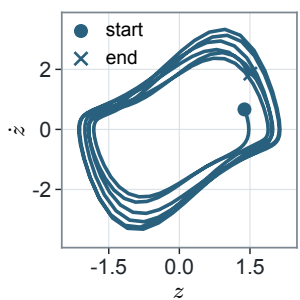
a Forced Van der Pol Oscillator

Identified equation:

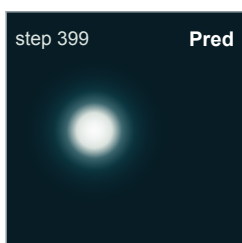
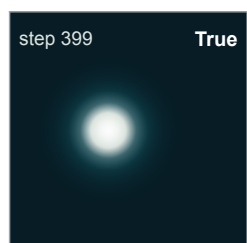
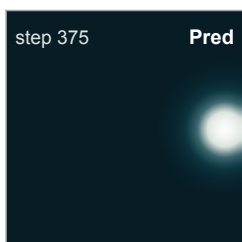
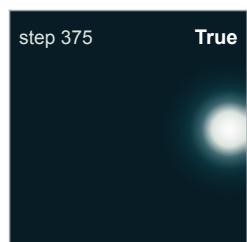
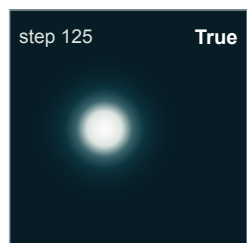
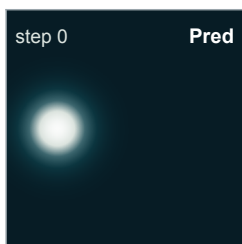
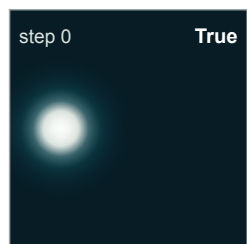
$$\ddot{z} = -1.0z + 1.2\dot{z} - 1.2z^2\dot{z} - 1.0f$$

True equation phase

Identified equation phase



Oscillator State Snapshots

**b Forced Cantilever Beam**

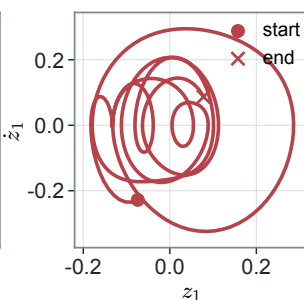
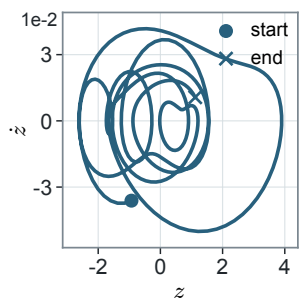
Identified equations:

$$\ddot{z}_1 = -10.1z_1 + 3.1\dot{z}_2 + 9.4f$$

$$\ddot{z}_2 = -1.0\dot{z}_1 - 2.8\dot{z}_2$$

True first-mode phase

Identified equation phase



Beam Deflection Snapshots

**c****Forced Cylinder Wake**

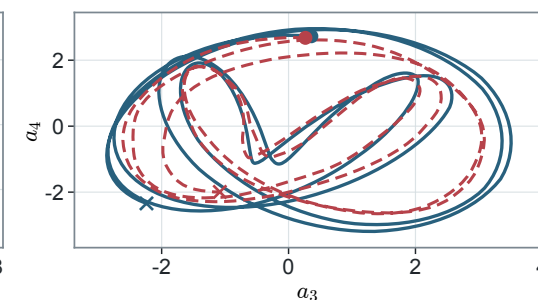
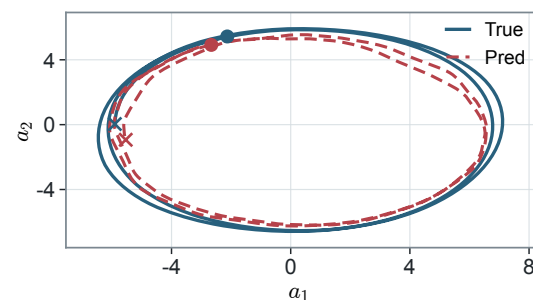
Identified equations:

$$\dot{z}_1 = -0.145z_1 - 0.022z_2 + 0.137z_3 - 0.008z_4 + 0.010z_2z_3 + 0.286\tilde{f}, \quad \dot{z}_2 = 0.009z_4$$

$$\dot{z}_3 = 0.130z_1 + 0.019z_2 - 0.124z_3 + 0.008z_4 - 0.010z_2z_3 - 0.258\tilde{f}, \quad \dot{z}_4 = -0.006z_2$$

Wake-field POD phase: modes 1-2

Wake-field POD phase: modes 3-4



Wake-Field Snapshots

